Cancer Detection & Classification Using Machine Learning

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Abstract

Early and accurate cancer diagnosis via histopathological image analysis is critical but remains challenging due to the complexity of tissue morphology. In this study, we develop and evaluate a deep learning-based classifier to distinguish between benign and malignant lung and colon tissues using the LC25000 dataset. Our approach compares a custom CNN and a pretrained ResNet-18 model under two training setups: a one-shot baseline and a rigorously preprocessed pipeline incorporating deep feature clustering, data split integrity, and Optuna-based hyperparameter tuning. The ResNet-18 model achieved 97.12% accuracy after full preprocessing, outperforming the one-shot counterpart (96.85%). Our results demonstrate the importance of robust preprocessing for generalizable medical AI and emphasize the risks of overestimating performance due to improper data handling. The proposed framework highlights key steps toward building reproducible, clinically relevant diagnostic models.

1 Introduction

Early detection of cancer is critical for improving patient outcomes, yet diagnosing cancer through histopathological image analysis is a challenging task due to the complexity and variability of tissue samples. Traditional manual analysis by experts can be time-consuming and subject to variability, motivating the need for automated diagnostic tools [1], [2].

In this project, we develop a deep learning-based classifier to distinguish between healthy and cancerous tissues in lung and colon samples. Our approach leverages state-of-the-art neural network architectures, including a custom Convolutional Neural Network (CNN) and a fine-tuned ResNet-18 model, to improve diagnostic accuracy and reliability.

The dataset employed in this study originates from the LC25000 collection as described by Borkowski et al. [3]. The original histopathological images were acquired from pathology glass slides, consisting of 750 lung tissue images (250 benign lung tissue, 250 lung adenocarcinomas, and 250 lung squamous cell carcinomas) and 500 colon tissue images (250 benign colon tissue and 250 colon adenocarcinomas). The images were cropped to a uniform resolution of 768 x 768 pixels and subsequently augmented using the Augmentor software. This augmentation process—applying transformations such as rotations (up to 25 degrees) and horizontal and vertical flips—expanded the dataset to a total of 25,000 color images, with 5,000 images per class.

2 Methodology

2.1 Preprocessing

To ensure robust evaluation and prevent data leakage, a rigorous preprocessing pipeline was developed for the LC25000 dataset. This pipeline was designed to be reproducible, scalable, and suitable for both full and reduced datasets, depending on the experimental setting.

We first standardized the environment by setting random seeds for NumPy and PyTorch to enable reproducibility. The code was configured to automatically use GPU acceleration when available, thus ensuring scalability for large datasets. All file paths and configurations were modularized to support both full and sample datasets through a toggleable flag.

For feature extraction, we employed transfer learning using a pretrained ResNet-18 model from the PyTorch library [4, 5]. The final layer of the network was removed, transforming the model into a fixed feature extractor. Input images were resized, normalized, and processed in batches according to ResNet-18 requirements, to efficiently extract deep features representing high-level image characteristics.

To structure the dataset and avoid potential data leakage, unsupervised clustering was applied to the extracted features using the K-Means algorithm. The number of clusters was selected based on the number of original samples per class (250 clusters for each class), and cluster centroids were used to group semantically similar images.

The clustered images were then split into training, validation, and test sets (70%/15%/15%) while ensuring cluster integrity within each split. This clustering-based splitting approach guarantees that semantically similar images do not appear across multiple splits, effectively mitigating data leakage.

Finally, the data was copied into separate directories for each split to align with standard deep learning training pipelines. Visualizations were generated to inspect the distribution of images across classes and splits, verifying the balance and correctness of the final dataset.

2.2 Models

Two distinct deep learning architectures were employed, a simple convolutional neural network (CNN) and a pretrained ResNet-18 model [4]. The simple CNN served as a lightweight, interpretable baseline with a reduced number of parameters. It comprised multiple convolutional layers with ReLU activation and max-pooling, followed by fully connected layers for classification. For a complete layer-by-layer description and the full implementation code, see Appendix A.

In contrast, the ResNet-18 architecture, pretrained on ImageNet, leveraged transfer learning by fine-tuning its final layers on the target medical image dataset [4]. This approach allowed the model to benefit from previously learned visual features while adapting to the new domain-specific task. The final classification head was replaced with a fully connected layer to match the number of classes in the dataset (5 classes). Both models were implemented using PyTorch, with support for GPU acceleration when available.

2.3 Training

For training, the preprocessed and split dataset, as described in the previous section 2.1, was utilized. Hyperparameter optimization for both models was conducted using Optuna, with the goal of maximizing validation accuracy [6]. The hyperparameters learning rate and batch size were explored over 10 epochs and 10 trials, and the final models were retrained on the full training set using the best set of hyperparameters from the trials. The training and validation loss and accuracy can be seen in the Appendix B.

To explore the data leakage from a non-preprocessed dataset, a one-shot training approach was implemented. In this setup, the same CNN and ResNet-18 were trained with fixed parameters —using a batch size of 32 and a learning rate of 0.001— for one epoch. Additionally, instead of applying the previously described preprocessing and

Fig. 1 Histogram illustrating the distribution of images per cluster for the *colon adenocarcinomas* class.

cluster splitting (see 2.1), the original data from LC25000 was directly partitioned into test, validation, and training sets.

3 Results

3.1 Preprocessing Results

Clustering Outcomes

As part of the preprocessing pipeline, K-Means clustering was applied to the deep features extracted via the pretrained ResNet-18. Figure 1 shows the distribution of images per cluster for the *colon adenocarcinomas* class. The smallest cluster contains 3 images and the largest cluster 61 images. The average cluster size was 20, and the median 19. These statistics indicate that while most clusters center around 20 images, there is still noticeable variability in cluster size.

Data Split Analysis

Following the clustering process, the clustered data were partitioned into training, validation, and testing sets. Table 1 summarizes the total number of images for each set and the distribution across the different classes. Ideally, one would aim for a 70%/15%/15% split; however, due to the necessity of preserving cluster integrity, the actual split is slightly different. Out of a total of 25,000 images, the training set comprises 17,464 (69.86%), the validation set 3,737 (14.95%), and the testing set 3,799 images (15.20%).

Table 1 Distribution of Images per Class in Training, Validation, and Testing Sets

Class / Set	Training	Validation	Testing
Colon Adenocarcinoma	3533 (70.66%)	721 (14.42%)	746 (14.92%)
Colon Benign Tissue	3632 (72.64%)	710 (14.20%)	658 (13.16%)
Lung Adenocarcinoma	3488 (69.76%)	797 (15.94%)	715 (14.30%)
Lung Squamous Cell Carcinoma	3334 (66.68%)	758 (15.16%)	908 (18.16%)
Lung Benign Tissue	3477 (69.54%)	751 (15.02%)	772 (15.44%)
Total	17464 (69.86%)	3737 (14.95%)	3799 (15.20%)

3.2 Model Results

To assess the proposed approaches, see 2.3, we conducted two experimental setups: one using a one-shot training scenario—without clustering or hyperparameter tuning—and another employing full preprocessing with cluster-based splitting and Optuna-driven hyperparameter optimization. Table 2 summarizes the final accuracies for the ResNet-18 and CNN models under both conditions. Additionally, Table 3 provides a detailed classification report for the fully preprocessed models, including precision, recall, and F1-scores for each class.

Table 2 Comparison of Model Accuracies: Preprocessed & Tuned vs. One-Shot Approaches

Model	Preprocessed & Tuned	One-Shot (No Preprocessing)
ResNet-18	97.12%	96.85%
CNN	91.03%	90.80%

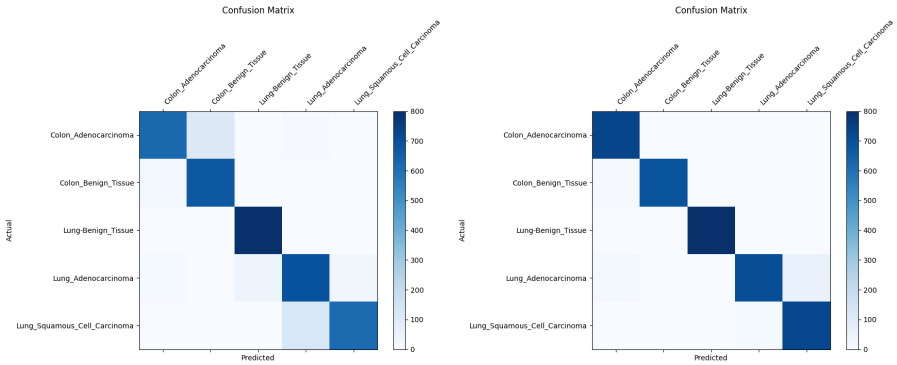

Fig. 2 Confusion matrix for test set with CNN (left) and ResNet-18 (right) for the preprocessed date and after finetuning.

Specifically, the ResNet-18 model achieved an accuracy of 97.12% with full preprocessing, compared to 96.85% in the one-shot scenario. Likewise, the CNN reached accuracies of 91.03% and 90.80%, respectively. The class-level metrics in Table 3 confirm that both models perform consistently across all classes, with only minor variations in precision, recall, and F1-scores.

Table 3 Classification Report for the Fully Preprocessed Models

Class	ResNet-18			CNN		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score
Colon Adenocarcinoma	0.97	1.00	0.98	0.95	0.85	0.89
Colon Benign Tissue	1.00	0.99	0.99	0.87	0.97	0.92
Lung Benign Tissue	1.00	1.00	1.00	0.95	1.00	0.97
Lung Adenocarcinoma	0.98	0.91	0.95	0.84	0.90	0.87
Lung Squamous Cell Carcinoma	0.93	0.98	0.95	0.96	0.83	0.89

As can be seen in Figure 2, both fully trained models were able to distinguish between colon and lung tissue. The confusion matrices of the one-shot approach can be seen in the Appendix C.

4 Discussion

This study presents a preprocessing-aware deep learning pipeline for histopathological image classification, addressing common pitfalls such as data leakage and distributional bias. The results indicate that even relatively small adjustments in data preparation and model optimization can lead to measurable performance gains—especially in sensitive domains like medical imaging.

4.1 Clustering and Data Split Integrity

The clustering of deep features prior to dataset splitting was a critical step in mitigating identity leakage and overfitting. As highlighted by Yagis et al. [7] and Tampu et al. [8], conventional random splitting can lead to data leakage, particularly in medical image datasets where similar or sequential slices may appear in multiple sets. By

applying clustering before splitting, this study aligns with best practices and ensures stronger generalizability of the trained models.

The observed variability in cluster sizes (ranging from 3 to 61 images) indicates a natural heterogeneity in the feature space, which is consistent with prior observations of domain shift in histopathology datasets [9]. Nonetheless, the final data split maintained a balanced distribution across all classes, ensuring a fair evaluation of model performance.

4.2 Model Performance and Preprocessing Impact

The comparative evaluation demonstrated that preprocessing—including cluster-based splitting and hyperparameter optimization using Optuna [6]—yielded a consistent improvement in classification accuracy. Specifically, the ResNet-18 model's performance improved from 96.85% to 97.12%, and the CNN from 90.80% to 91.03%. While these improvements appear modest, they are meaningful in the context of clinical applications, where small gains can influence patient outcomes [1, 10].

Additionally, the class-level metrics indicate high and consistent precision, recall, and F1-scores across all diagnostic categories. These findings confirm the robustness of the pipeline, particularly in distinguishing between malignant and benign tissues in both colon and lung samples.

4.3 Implications for Medical AI Development

The demonstrated effectiveness of preprocessing reinforces the importance of methodological rigor in medical AI. This aligns with recent findings by Batchkala et al. [11], who advocate for using clustering to curate high-quality, non-redundant datasets. It also supports calls by Varoquaux and Cheplygina [12] for reproducibility and data-centric evaluation in clinical machine learning.

Moreover, the results provide empirical support for Tampu et al.'s [8] argument that improper dataset handling can artificially inflate performance. By avoiding such pitfalls and demonstrating them using the one-shot approach, this study contributes a reproducible and transparent framework for future medical image analysis research.

4.4 Limitations and Future Work

Despite the encouraging results, several limitations should be noted. First, while clustering reduces the risk of data leakage, it may alter the natural class distribution or ignore subtle inter-class relationships. Second, although ResNet-18 performed strongly, future experiments could explore transformer-based models or self-supervised techniques, such as those discussed in FASNet [2].

Furthermore, evaluation in this work was limited to accuracy and per-class performance metrics. Future research could include calibration analysis and external validation to further assess generalization and clinical utility.

To improve the results further research should be conducted using the following as a guide.

- **Dataset Expansion and Diversity:** Acquiring a larger volume of unaugmented, high-quality data -ideally from multiple centers- will not only mitigate overfitting but also improve model generalizability.
- **Enhanced Augmentation Strategies:** Incorporating more sophisticated augmentation techniques (e.g., elastic deformations, color jitter, or adversarial augmentations) may help simulate real-world variability and further reduce overfitting similar to Tellez et al. [13].
- **Robust Data Splitting Protocols:** Emphasizing patient-level or case-level splitting before augmentation remains critical to ensure that evaluation metrics reflect true generalization capabilities.
- **Exploration of Advanced Architectures:** Investigating newer network architectures or hybrid models could offer further improvements in performance and interpretability, which is vital for clinical integration.

- **Explainability and Interpretability:** Future work should also focus on incorporating model interpretability techniques to help clinicians understand the basis of the automated decisions, thereby fostering trust in the technology.

5 Conclusion & Further Research

In conclusion, our study demonstrates that deep learning classifiers -specifically, a custom-designed CNN and a fine-tuned ResNet-18- can achieve high accuracy in detecting and classifying cancerous tissues. With ResNet-18 achieving an accuracy exceeding 97% and the CNN surpassing 91%, our results underscore the potential of these models to enhance diagnostic workflows and support automated cancer screening. Models, like ours, could help to reduce the stress medical staff is under by reducing their workload while maintaining high quality diagnosis [14].

A central lesson from this work is the paramount importance of rigorous data management similar to Batchkala et al. [11]. Our findings highlight that proper data splitting -performed prior to any augmentation- is essential to prevent data leakage. Such leakage can falsely inflate performance metrics by allowing the model to "memorize" rather than generalize, thereby undermining clinical applicability.

Despite the promising performance metrics, our experiments also revealed significant challenges. The dataset, while valuable, presented limitations related to inherent leakage risks and the narrow scope of the applied augmentation techniques (limited to rotations and flips). This constrained variability may contribute to overfitting, indicating that the current data augmentation strategy does not fully capture the natural heterogeneity present in medical imaging data.

Overall, our work reinforces that meticulous data management and methodological rigor are not only key to achieving high diagnostic performance but are also essential for the development of reliable and clinically relevant automated diagnostic systems. By addressing current limitations and exploring advanced strategies, we can pave the way for robust, generalizable models that ultimately contribute to improved patient outcomes in cancer detection.

Appendix

A Model Architecture Details and Code

Below is the complete implementation of the simple CNN used in our experiments.

Listing 1 SimpleCNN Implementation

```
import torch.nn as nn

class SimpleCNN(nn.Module):
    def __init__(self, num_classes=5):
        super(SimpleCNN, self).__init__()
        # Convolutional layers to extract features from images
        self.features = nn.Sequential(
            nn.Conv2d(3, 16, kernel_size=3, stride=1, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(2, 2),

            nn.Conv2d(16, 32, kernel_size=3, stride=1, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(2, 2),

            nn.Conv2d(32, 64, kernel_size=3, stride=1, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(2, 2)
        )
        # Fully connected layers for classification
        self.classifier = nn.Sequential(
```

```
            nn.Linear(64 * 32 * 32, 128),
            nn.ReLU(),
            nn.Dropout(0.5),
            nn.Linear(128, num_classes)
        )

    def forward(self, x):
        x = self.features(x)
        x = x.view(x.size(0), -1)  # Flatten the feature maps
        x = self.classifier(x)
        return x
```

B Loss & Accuracy During Training

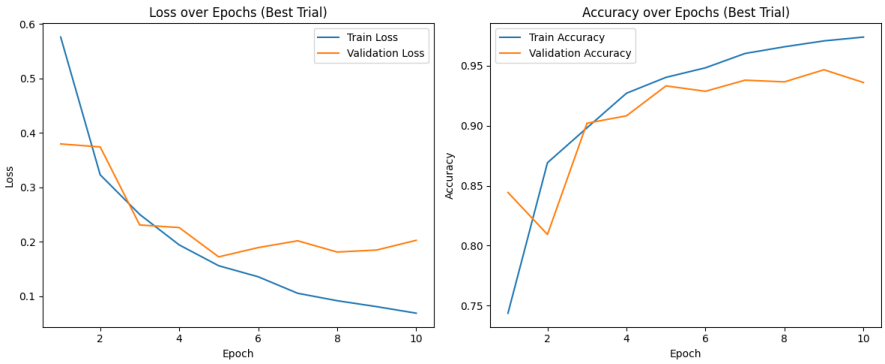

Fig. 3 Training and validation loss and accuracy for CNN during training with best hyperparameter set.

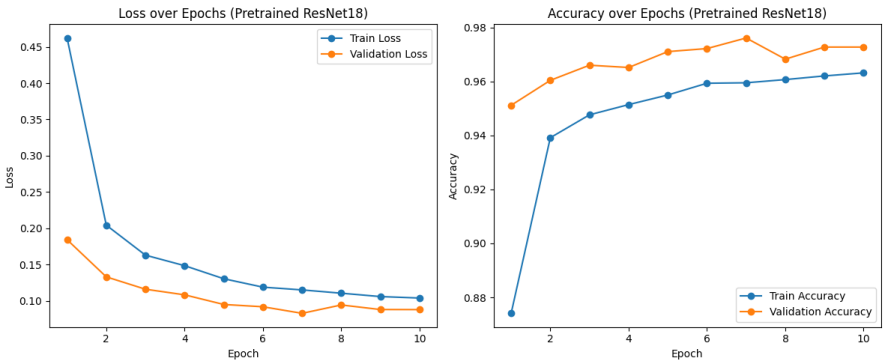

Fig. 4 Training and validation loss and accuracy for ResNet-18 during training with best hyperparameter set.

C Confusion Matrices for One-Shot Approach

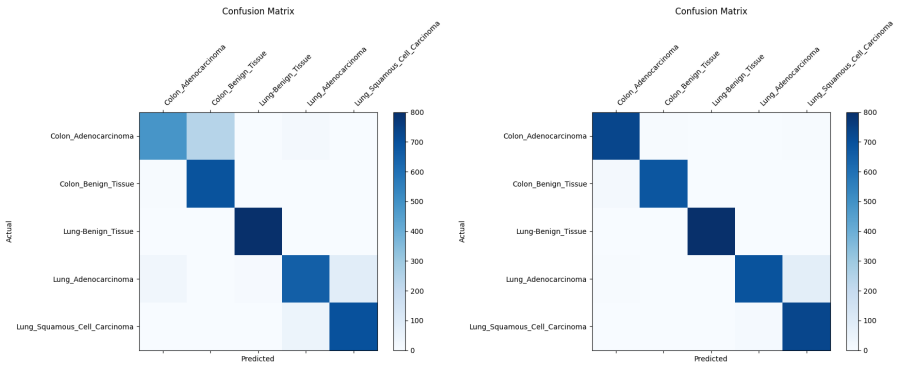

Fig. 5 Confusion matrix for test set with CNN (left) and ResNet-18 (right) for the one-shot approach with no data preprocessing.

D Use of Generative AI

Machine learning tools were utilized in various phases of this work. Specifically, ChatGPT was employed for text and code corrections, literature research, and brief summaries. The use of these tools is documented here in accordance with the guidelines of the ZHAW School of Engineering.

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